A Composite Bristle–Mindlin Friction Model for Mecanum Wheel Dynamics under Adaptive Sliding-Mode Control

Synthesis of Adamov Multicomponent Friction, LuGre Pre-Sliding Dynamics, and Mindlin Partial-Slip Correction (Revision 2)

Abstract

Standard friction models for omnidirectional mecanum platforms — most commonly the Adamov multicomponent friction law — assume gross sliding at every contact and integrate Coulomb friction over a Hertzian patch under combined translation and spin. When closed-loop adaptive sliding-mode controllers drive the platform into the microslip regime, the slip velocity at the rollers falls below the Stribeck transition and Adamov's assumption is violated over the majority of the trajectory. This work proposes a composite model that retains Adamov's kinematic slip-velocity decomposition at each roller but replaces the gross-sliding force law with a two-dimensional LuGre bristle model whose stiffness and damping coefficients are calibrated from Hertz–Mindlin contact mechanics for polyurethane rollers. Spin–translation coupling is preserved through the pressure-weighted mean local slip velocity, and the gross-sliding bias of this coupling is corrected by a state-based Mindlin slip-annulus ramp. The formulation is C^1 continuous, recovers Adamov exactly at gross sliding, reduces to standard LuGre at full stick, and is dimensionally consistent across both the translational and rotational channels.

1. Introduction

The multicomponent friction model of Adamov and Saypulaev [1], [2] extends Coulomb friction to a circular Hertzian contact patch under combined translation and spin, yielding closed-form expressions for the planar friction forces and the spin friction torque. This has become a standard formulation for KUKA youBot dynamics. However, it is derived under full gross sliding, and its validity in the microslip regime — precisely where closed-loop adaptive sliding-mode control [3] drives the slip velocity — is limited.

The LuGre dynamic friction model [4], [5] captures pre-sliding elastic deflection and the transition to gross sliding through a bristle state. Its two-dimensional tire extensions [6], [7] and the unifying framework of Pfitzer et al. [8] establish that bristle models faithfully represent directionally coupled friction, but they do not incorporate contact-patch spin, which is essential for mecanum rollers.

We propose a composite model integrating: (i) Adamov's kinematic slip-velocity computation [1], [2]; (ii) two-dimensional LuGre bristle dynamics for translational friction [7], [8] and a scalar LuGre formulation for spin friction; (iii) Hertz–Mindlin contact mechanics [9], [10] for the bristle coefficients; (iv) the Adamov pressure-weighted mean local slip [1], [11] as a shared coupling term; and (v) a state-based Mindlin slip-annulus correction ramp [9], [12]. Section 3 establishes the C^1 continuity and friction-balance properties that make the composite suitable for stiff implicit integration.

2. Proposed Composite Friction Model

Dimensional convention. Throughout, $\sigma_0 = 8 \cdot G^* \cdot a$ is the Mindlin tangential stiffness in N/m, and $\sigma_{0,rot} = (16/3) \cdot G^* \cdot a^3$ is the torsional stiffness in N·m/rad. The Stribeck functions are friction *capacities*: $g_{xy} = \mu(V_{eff}) \cdot N$ is a force [N], and $g_{mz} = \mu(V_{eff}) \cdot N \cdot a$ is a torque [N·m], where $\mu(V) = f_c + (f_s - f_c) \cdot \exp[-(V/v_s)^\alpha]$. With this convention the friction force and torque follow directly as $F = \sigma_0 \cdot z + \sigma_1 \cdot \dot{z}$ and $M_z = \sigma_{0,rot} \cdot z_{rot} + \sigma_{1,rot} \cdot \dot{z}_{rot}$ — with *no* additional normal-force factor, since N is already embedded in σ_0 (through a) and in the capacity g.

2.1 Kinematic Slip Velocity at Each Roller

The kinematic decomposition of the slip velocity at roller i, retained from Adamov [1]:

$$V_{PiX} = V_X - \rho_{Yi}\cdot\Omega - \dot{\varphi}_i\cdot R + \dot{\gamma}_i\cdot\sin(\delta_i)\cdot(R_d\cdot\cos(\tilde{\varphi}_i) - R) - \Delta Y_i\cdot(\Omega - \dot{\gamma}_i\cdot\sin(\tilde{\varphi}_i)\cdot\cos(\delta_i)) \quad (1a)$$

$$V_{PiY} = V_Y + \rho_{Xi}\cdot\Omega + \dot{\gamma}_i\cdot\cos(\delta_i)\cdot(R\cdot\cos(\tilde{\varphi}_i) - R_d) \quad (1b)$$

$$\omega_{zi} = \Omega - \dot{\gamma}_i\cdot\sin(\tilde{\varphi}_i)\cdot\cos(\delta_i) \quad (1c)$$

2.2 Hertz–Mindlin Contact Coefficients (precomputed per wheel)

Under the static normal force assumption [1], all coefficients are computed once at construction from the static normal force N_i:

$$a_i = (\ 3\cdot N_i\cdot R_{roller} / (4\cdot E^*)\ )^{1/3} \quad [m] \quad (2)$$

$$\sigma_{0,i} = 8\cdot G^*\cdot a_i \quad [N/m] \quad (3)$$

$$\sigma_{0,rot,i} = (16/3)\cdot G^*\cdot a_i^3 \quad [N\cdot m/rad] \quad (4)$$

$$\sigma_{1,i} = 2\cdot\zeta\cdot\sqrt{}(\sigma_{0,i}\cdot m_{eff}) \quad [N\cdot s/m] \quad (5)$$

$$\sigma_{1,rot,i} = 2\cdot\zeta\cdot\sqrt{}(\sigma_{0,rot,i}\cdot J_{eff}) \quad [N\cdot m\cdot s/rad] \quad (6)$$

$$\delta^*_i = f_s\cdot N_i / \sigma_{0,i} \quad [m] \quad (7)$$

$$\delta^*_{rot,i} = f_s\cdot N_i\cdot a_i / \sigma_{0,rot,i} \quad [rad] \quad (8)$$

2.3 Mindlin Slip-Annulus Fraction (State-Based Ramp)

The Adamov coupling coefficients were derived for full gross sliding and overestimate spin–translation coupling in the microslip regime. The coupling is modulated by the Mindlin slip-annulus area fraction, expressed in terms of the normalized bristle deflection $\xi = |z|/\delta^*$:

$$\xi_{safe} = \xi / \sqrt{}(1 + \xi^2) \quad (9)$$

$$R(\xi) = 1 - (\ 1 - \xi_{safe}\ )^{2/3} \quad (10)$$

The saturator ξ_{safe} maps $[0, \infty) \to [0, 1)$, removing the derivative singularity of the bare Mindlin form at $\xi = 1$ and rendering the ramp C^∞ smooth.

2.4 Effective Slip Velocities (Adamov Pressure-Weighted Mean)

Let $\mathbf{V_p} = (V_{px}, V_{py})$ be the translational slip vector with norm $\|V_p\|$. The translational effective slip is in m/s; the rotational effective slip — obtained by dividing the Adamov Mz-denominator by the contact radius a — is in rad/s:

$$V_{eff,xy} = \|V_p\| + R_{xy}(\xi_{xy}) \cdot (8/(3\pi)) \cdot |\omega_z| \cdot a \quad [m/s] \quad (11)$$

Corrected rotational effective slip. The original Adamov Mz-denominator $(16/3\pi)\cdot|\omega_z|\cdot a + 5\cdot\|V_p\|$ has units of m/s. Dividing by a converts it to an angular rate, as required for dimensional consistency with the torsional bristle ODE (Eq. 17):

$$V_{eff,mz} = (16/(3\pi)) \cdot |\omega_z| + R_{mz}(\xi_{rot}) \cdot 5\cdot\|V_p\| / a \quad [rad/s] \quad (12)$$

Here $\xi_{xy} = \|\mathbf{z}_{xy}\|/\delta^*$ and $\xi_{rot} = |z_{rot}|/\delta^*_{rot}$. In each channel the native term (translation in Eq. 11, spin in Eq. 12) is unramped, while the cross-coupling term (spin in Eq. 11, translation in Eq. 12) is scaled by the Mindlin slip-annulus ramp of that channel.

2.5 Stribeck Friction Capacity

The capacities use the standard Stribeck law [4], [5] with rubber exponent $\alpha = 1/2$ [6], scaled to force (N) and torque (N·m) respectively:

$$g_{xy} = [\ f_c + (f_s - f_c)\cdot\exp(\ -(\ V_{eff,xy} / v_s\ )^\alpha\ )\ ] \cdot N \quad [N] \quad (13)$$

$$g_{mz} = [\, f_c + (f_s - f_c)\cdot\exp(\,-(\,V_{\mathrm{eff},mz} / v_{s,\mathrm{rot}}\,)^{\alpha}\,)\,] \cdot N \cdot a \quad [\mathrm{N\cdot m}] \tag{14}$$

2.6 Bristle State Dynamics

The mean slip speed appears in the numerator of the dissipation coefficient, consistent with [7], [8] and required for asymptotic Adamov recovery:

$$dz_x/dt = V_{px} - \sigma_0 \cdot V_{\mathrm{eff},xy} / g_{xy} \cdot z_x \quad [\mathrm{m/s}] \tag{15}$$

$$dz_y/dt = V_{py} - \sigma_0 \cdot V_{\mathrm{eff},xy} / g_{xy} \cdot z_y \quad [\mathrm{m/s}] \tag{16}$$

$$dz_{\mathrm{rot}}/dt = \omega_z - \sigma_{0,\mathrm{rot}} \cdot V_{\mathrm{eff},mz} / g_{mz} \cdot z_{\mathrm{rot}} \quad [\mathrm{rad/s}] \tag{17}$$

2.7 Friction Forces and Spin Torque

With the capacity convention of Section 2.5, the friction force and torque follow directly, with no additional normal-force factor:

$$F_x = \sigma_0\cdot z_x + \sigma_1\cdot dz_x/dt \quad [\mathrm{N}] \tag{18}$$

$$F_y = \sigma_0\cdot z_y + \sigma_1\cdot dz_y/dt \quad [\mathrm{N}] \tag{19}$$

$$M_z = \sigma_{0,\mathrm{rot}}\cdot z_{\mathrm{rot}} + \sigma_{1,\mathrm{rot}}\cdot dz_{\mathrm{rot}}/dt \quad [\mathrm{N\cdot m}] \tag{20}$$

2.8 Asymptotic Recovery of Adamov

At gross sliding (R → 1, z → δ*), the steady-state bristle deflection $z_{ss} = g\cdot V_p/(\sigma_0\cdot V_{\mathrm{eff}})$ gives, for the two channels:

$$F_{ss} = -f_c\cdot N \cdot V_p / (\,\|V_p\| + (8/(3\pi))\cdot|\omega_z|\cdot a\,) \tag{21a}$$

$$M_{z,ss} = -f_c\cdot N\cdot a^2\cdot\omega_z / (\,(16/(3\pi))\cdot|\omega_z|\cdot a + 5\cdot\|V_p\|\,) \tag{21b}$$

These are exactly the Adamov multicomponent force and torque [1], [2]. The division by *a* in Eq. 12 and the torque capacity $g_{mz} = \mu\cdot N\cdot a$ in Eq. 14 combine so that the factor a^2 in Eq. 21b emerges correctly, confirming the dimensional consistency of the corrected rotational channel.

2.9 Augmented ODE State Vector

Three bristle states per roller ($z_{x,i}$, $z_{y,i}$, $z_{\mathrm{rot},i}$) for i = 1,…,4 augment the original 21-dimensional youBot state by 12, yielding 33 states. All friction quantities are smooth functions of the augmented state and the precomputed coefficients, ensuring compatibility with the fully implicit RadauIIA5 solver.

3. Continuity and Friction-Balance Properties

3.1 C¹ Continuity of the Adamov Coupling

The bare two-dimensional Coulomb law $F = -f\cdot N\cdot V_p/\|V_p\|$ is discontinuous at $V_p = 0$: the direction is undefined and the magnitude jumps from f·N to 0. The Adamov denominator removes this singularity whenever spin is present. With $\omega_z \neq 0$, the denominator is bounded below by a strictly positive quantity:

$$\|V_p\| + (8/(3\pi))\cdot|\omega_z|\cdot a \geq (8/(3\pi))\cdot|\omega_z|\cdot a > 0 \tag{22}$$

so the Adamov force $F = -f\cdot N\cdot V_p/(\|V_p\| + (8/3\pi)\cdot|\omega_z|\cdot a)$ is a smooth (C^{∞}) function of V_p through $V_p = 0$. Physically, contact-patch spin keeps every surface element sliding even when the patch centre has zero translational slip, so there is no breakaway discontinuity. The spin term acts as a built-in regularizer for the translational Coulomb singularity — a regularization with a mechanical origin rather than the numerical $\tanh(k\cdot\|V_p\|)$ smoothing used in the baseline simulator.

The single remaining non-smooth point is the simultaneous corner $V_p = 0$ and $\omega_z = 0$, where the contact is fully stuck. The LuGre composite removes even this: the friction force $F = \sigma_0 \cdot z + \sigma_1 \cdot \dot{z}$ is a smooth function of the bristle state z, which evolves continuously through the ODE (Eqs. 15–17). At a velocity reversal, z passes smoothly through zero rather than switching sign discontinuously, so F and M_z remain C^1 (indeed C^∞) everywhere, including the full-stick corner. The composite is therefore globally C^1 — a strict improvement over the Adamov-plus-tanh formulation, which is only C^0 at the corner and relies on a numerically tuned smoothing width.

3.2 Balance Between Spin and Translational Friction

The shared Adamov denominators encode the competition between translation and spin for the contact's finite friction budget. Each surface element obeys the pointwise Coulomb bound $|\tau(\mathbf{r})| \leq \mu \cdot p(\mathbf{r})$; the integrated force and torque therefore cannot both reach their maxima simultaneously. The two denominators express this trade-off:

$$D_{xy} = \|V_p\| + (8/(3\pi)) \cdot |\omega_z| \cdot a \tag{23a}$$

$$D_{mz} = 5 \cdot \|V_p\| + (16/(3\pi)) \cdot |\omega_z| \cdot a \tag{23b}$$

In the translation-dominant limit ($\|V_p\| \gg |\omega_z| \cdot a$), the force recovers the full Coulomb magnitude $\|F\| \to f \cdot N$ while the spin torque is suppressed as $M_z \to -f \cdot N \cdot a^2 \cdot \omega_z / (5 \cdot \|V_p\|) \to 0$. In the spin-dominant limit ($|\omega_z| \cdot a \gg \|V_p\|$), the force is suppressed as $\|F\| \to f \cdot N \cdot \|V_p\| / ((8/3\pi) \cdot |\omega_z| \cdot a) \to 0$ while the torque saturates at its pure-spin maximum $|M_z| \to (3\pi/16) \cdot f \cdot N \cdot a$. The total frictional power is conserved in the sense that dissipation lost by one channel is gained by the other.

This balance is made explicit by the dissipation rate, which sums the two channels and is non-negative throughout:

$$P_{diss} = g_{xy} \cdot \|V_p\|^2 / V_{eff,xy} + g_{mz} \cdot \omega_z^{\,2} / V_{eff,mz} \geq 0 \tag{24}$$

The coefficient asymmetry between the channels — (8/3π) versus (16/3π), and the factor 5 multiplying $\|V_p\|$ in D_{mz} — arises from the moment-arm weighting in the contact integral. The torque integrand carries an extra factor of radial distance, so it weights the outer annulus of the patch more heavily, where the spin-induced local slip velocity is largest. This is why spin saturates the torque channel before it saturates the force channel, and is the mechanical origin of the partial-slip coupling the state-based ramp (Section 2.3) restores in the microslip regime.

4. Discussion

Each constituent of the composite has a well-established ancestor — the LuGre bristle formulation [4], [5], its 2D extensions [6], [7], [8], Hertz–Mindlin contact mechanics [9], [10], the Zhuravlev–Adamov integration of Coulomb friction under combined sliding and spin [1], [2], [11], and Mindlin partial-slip theory [9], [12]. The novelty lies in their composition: to the best of our knowledge no prior formulation (a) uses the Adamov pressure-weighted mean local slip as the dissipation driver of a 2D LuGre bristle ODE for a mecanum roller, (b) applies a state-based Mindlin slip-annulus ramp to correct the gross-sliding coupling bias in the microslip regime, or (c) resolves the dimensional consistency of the rotational channel through normalization of the effective slip by the Hertz contact radius.

The motivation arises from the operational regime of adaptive sliding-mode controllers on omnidirectional platforms [3], which structurally drive the contact into pre-sliding where the Adamov gross-sliding assumption fails. The PINN literature for friction [13], [14], [15], [16] has begun to incorporate LuGre-style physics losses but has not addressed training-data quality under closed-loop microslip on mecanum platforms. The proposed model serves both as a higher-fidelity simulator for data generation and as a candidate physics-residual structure for the training loss.

Limitations include the static normal force assumption [1], the bounded overestimation of pre-sliding stiffness under combined loading (Mindlin assumption A3 [9]), and the use of gross-sliding–derived coupling coefficients exact only asymptotically. The state-based ramp bounds these errors and forces them to vanish in both the full-stick and gross-sliding limits, but the composite remains an approximation of the full distributed bristle field [17], [18] in the intermediate regime.

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Appendix A. LaTeX Source for Microsoft Word Equation Editor

Each equation below is a standalone expression in Word-compatible LaTeX linear format. To use: in Word, press `Alt + =` to insert an equation, set the input mode to *LaTeX* on the Equation tab, then paste a single block and press space (or Enter) to convert it to a professional equation. The blocks avoid `align` / `tag` environments (which Word does not parse) and use only commands the Word editor supports: `\frac`, `\sqrt`, `\dot`, `\|`, Greek letters, `\cdot`, and sub/superscripts. Multi-term equations are kept on one line; long ones may wrap in Word but remain a single equation object. Subscript words such as `eff` may render italic; switch them to upright via Word's formatting if desired.

Eq. (1a) — slip velocity, X component

```
V_{PiX} = V_X - \rho_{Yi}\Omega - \dot{\varphi}_i R
  + \dot{\gamma}_i \sin\delta_i (R_d \cos\tilde{\varphi}_i - R)
  - \Delta Y_i (\Omega - \dot{\gamma}_i \sin\tilde{\varphi}_i \cos\delta_i)
```

Eq. (1b) — slip velocity, Y component

```
V_{PiY} = V_Y + \rho_{Xi}\Omega + \dot{\gamma}_i \cos\delta_i (R \cos\tilde{\varphi}_i - R_d)
```

Eq. (1c) — contact spin velocity

```
\omega_{zi} = \Omega - \dot{\gamma}_i \sin\tilde{\varphi}_i \cos\delta_i
```

Eq. (2) — Hertz contact radius

```
a_i = \left( \frac{3 N_i R_{roller}}{4 E^*} \right)^{1/3}
```

Eq. (3) — Mindlin tangential stiffness

```
\sigma_{0,i} = 8 G^* a_i
```

Eq. (4) — Mindlin torsional stiffness

```
\sigma_{0,rot,i} = \frac{16}{3} G^* a_i^3
```

Eq. (5) — translational bristle damping

```
\sigma_{1,i} = 2 \zeta \sqrt{\sigma_{0,i} m_{eff}}
```

Eq. (6) — rotational bristle damping

```
\sigma_{1,rot,i} = 2 \zeta \sqrt{\sigma_{0,rot,i} J_{eff}}
```

Eq. (7) — translational pre-sliding distance

```
\delta_i^* = \frac{f_s N_i}{\sigma_{0,i}}
```

Eq. (8) — rotational pre-sliding distance

```
\delta_{rot,i}^* = \frac{f_s N_i a_i}{\sigma_{0,rot,i}}
```

Eq. (9) — smooth saturator

```
\xi_{safe} = \frac{\xi}{\sqrt{1 + \xi^2}}
```

Eq. (10) — Mindlin slip-annulus ramp

```
R(\xi) = 1 - (1 - \xi_{safe})^{2/3}
```

Eq. (11) — translational effective slip [m/s]

```
V_{eff,xy} = \|V_p\| + R_{xy} \cdot \frac{8}{3\pi} \cdot |\omega_z| \cdot a
```

Eq. (12) — rotational effective slip [rad/s] (corrected)

```
V_{eff,mz} = \frac{16}{3\pi} \cdot |\omega_z| + R_{mz} \cdot \frac{5 \|V_p\|}{a}
```

Eq. (13) — translational Stribeck capacity [N]

```
g_{xy} = \left[ f_c + (f_s - f_c)
  \exp\left( -\left( \frac{V_{eff,xy}}{v_s} \right)^{\alpha} \right) \right] N
```

Eq. (14) — rotational Stribeck capacity [N·m]

```
g_{mz} = \left[ f_c + (f_s - f_c)
  \exp\left( -\left( \frac{V_{eff,mz}}{v_{s,rot}} \right)^{\alpha} \right) \right] N a
```

Eq. (15) — bristle ODE, x

```
\dot{z}_x = V_{px} - \sigma_0 \frac{V_{eff,xy}}{g_{xy}} z_x
```

Eq. (16) — bristle ODE, y

```
\dot{z}_y = V_{py} - \sigma_0 \frac{V_{eff,xy}}{g_{xy}} z_y
```

Eq. (17) — bristle ODE, rotational

```
\dot{z}_{rot} = \omega_z - \sigma_{0,rot} \frac{V_{eff,mz}}{g_{mz}} z_{rot}
```

Eq. (18) — friction force, x [N]

```
F_x = \sigma_0 z_x + \sigma_1 \dot{z}_x
```

Eq. (19) — friction force, y [N]

```
F_y = \sigma_0 z_y + \sigma_1 \dot{z}_y
```

Eq. (20) — spin friction torque [N·m]

```
M_z = \sigma_{0,rot} z_{rot} + \sigma_{1,rot} \dot{z}_{rot}
```

Eq. (21a) — gross-sliding force (Adamov recovery)

```
F_{ss} = -\frac{f_c N V_p}{\|V_p\| + \frac{8}{3\pi} |\omega_z| a}
```

Eq. (21b) — gross-sliding torque (Adamov recovery)

```
M_{z,ss} = -\frac{f_c N a^2 \omega_z}{\frac{16}{3\pi} |\omega_z| a + 5 \|V_p\|}
```

Eq. (22) — spin lower bound on the denominator

```
\|V_p\| + \frac{8}{3\pi} |\omega_z| a \geq \frac{8}{3\pi} |\omega_z| a > 0
```

Eq. (23a) — force-channel denominator

```
D_{xy} = \|V_p\| + \frac{8}{3\pi} |\omega_z| a
```

Eq. (23b) — torque-channel denominator

```
D_{mz} = 5 \|V_p\| + \frac{16}{3\pi} |\omega_z| a
```

Eq. (24) — total dissipation rate per contact

```
P_{diss} = \frac{g_{xy} \|V_p\|^2}{V_{eff,xy}} + \frac{g_{mz} \omega_z^2}{V_{eff,mz}} \geq 0
```

Note on norms. The token `\|` renders as the double-bar norm $\|\cdot\|$ in Word's equation editor. If a particular Word build does not recognize it, substitute `\left\| ... \right\|` or replace with single bars `|...|` and interpret as a vector magnitude.